

Mohamed El Baha

AI Research Engineer

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EXPERIENCE

Michelin | *Senior AI Engineer* | *Clermont-Ferrand, France*

Apr. 2022 - Present

- Lead the development of machine learning and deep learning solutions for industrial R&D applications, including generative design, predictive maintenance, time-series modelling, and reliability analysis.
- Design generative modelling pipelines using VAEs, diffusion models, and flow matching to generate and optimize engineering geometries under performance constraints.
- Develop predictive maintenance and survival-analysis models using sensor, operational, and maintenance data from industrial and mining applications.
- Build reusable Python libraries and production workflows to accelerate experimentation, model deployment, and adoption across engineering teams.
- Collaborate with simulation experts, performance analysts, data engineers, and product stakeholders to translate engineering requirements into deployable AI solutions.
- Deliver technical training on statistics, machine learning, survival analysis, and generative modelling to Michelin engineers and data scientists.

LIMSI | *Data Scientist* | *Paris, France*

Apr. 2021 – Nov 2021

- Implemented Machine learning algorithms related to Eye Gaze movement and speech, to predict the score of English tests. Deployed these algorithms via a Python interface.
- Conducted speech analysis using Hidden Markov Models, utilizing forced alignment to accurately synchronize speech and transcription data, providing valuable insight into language patterns and usage.

Orano TN | *Engineering Intern* | *Montigny le Bretonneux, France*

Sep. 2020 – Dec 2020

- Directed a team of five engineering students in conducting a comprehensive benchmark study to evaluate and enhance the efficiency of engineering processes.

EDUCATION

Executive Education in Leadership and Management

Saïd Business School, University of Oxford

- Leading Through Uncertainty and Disruption; Innovation for Business Leaders; Leadership Development

MEng in Mathematical and Computational Engineering

Top 10% of the class

IMT Atlantique (ex École des Mines x Telecom Bretagne)

- Selected coursework: Machine Learning, Deep Learning, Optimization, Operations Research, Numerical Methods, and Stochastic Modelling

BSc in Pure Mathematics – *Licence d'excellence*

Ranked 2nd of 13

University of Western Brittany

- Selected coursework: Advanced Calculus, Probability and Statistics, Complex Analysis, and Algebra

SKILLS

Languages: English (*Fluent*), French (*Bilingual*), Spanish (*Intermediate*), Arabic (*Native*)

Programming Languages: Python, Julia, R, Matlab

Tools & platforms: Git, GitLab, Kedro, Docker, Dataiku

Libraries & Frameworks Tensorflow, Keras, PyTorch, Pandas, HuggingFace, Numpy, Scikit, Seaborn

PUBLICATIONS

- *FlowCompose: Exact Multimodal Composition from Frozen Unimodal Flows*, F.Oubari, **M.El Baha**, A. Cherrat
- *GeMOVA : Generative Models Visual Atlas*, **M.EL Baha**, F. Oubari
- *Multi-Component VAE with Gaussian Markov Random Field*, Fouad Oubari, Mohamed **El Baha**, *AISTATS 2026*
- *A System for Automatic Calculation of the Eye-Voice Span*, **M. El Baha**, O. Augereau, S. Kobylanskaya, I. Vasilescu and L. Devillers, Document Analysis System 2021
- *Witt theory, a proof of Witt equivalence theorem*, **M. El Baha**, T. Levasseur, LMBA 2019

SIDE ACTIVITIES

- Founder & President of MAIR : Moroccans in AI Research, the largest Moroccan researcher in the AI community in Morocco.
- Executive of the MoroccoAI Committee: Non-profit Organization to Promote AI Growth in Morocco Across the Spectrum.
 - The Lead Organizer of InnovAI hackathon (+500 participants) December 2024.
 - The Lead Organizer of the GenAI hackathon (+150 participants) December 2023.
 - Mentor and organizer of AI Summer School (60+ participants) August 2021.
- Founder of SaveOurSeas : A local project for waste collections on Brittany seas, and organizing awareness events in Schools.

TEACHING

- Advanced Deep Learning - 5A : Polytech Clermont Ferrand - 2024-2025
- Statistics and data analysis for Tire Performance Analysts : 2023 -2024-2025

STUDENTS

- 2025 Shaimaa Abu Yousuf - Engineering Intern - Leukemia Detection Using Transfert Learning for Accurate Classification.
- 2024/2025 Rémy Vallot - PhD Candidate - ENS Paris Saclay - Solver Acceleration using ML methods
- 2023 Moad Yachouti - Engineering Intern - Université Paris Saclay - Explainable AI in Predictive ML models.

INTERESTS

AI Communities, Product x Design, Surf, Geopolitics